

## SAFETY DATA SHEET

Creation Date 07-May-1997

Revision Date 25-Dec-2021

Revision Number 4

### 1. Identification

**Product Name** Titanium(IV) ethoxide, contains 5-15% isopropanol

**Cat No. :** AC312540000; AC312540500; AC312542500

**Synonyms** Tetraethyl Orthotitanate.

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |            |
|--------------------------------------------------|------------|
| Flammable liquids                                | Category 3 |
| Serious Eye Damage/Eye Irritation                | Category 2 |
| Specific target organ toxicity (single exposure) | Category 3 |
| Target Organs - Central nervous system (CNS).    |            |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Flammable liquid and vapor  
Causes serious eye irritation  
May cause drowsiness or dizziness

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                  | CAS No    | Weight % |
|----------------------------|-----------|----------|
| Ethanol, titanium(4+) salt | 3087-36-3 | 85-95    |
| Isopropyl alcohol          | 67-63-0   | 5-15     |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 28 °C / 82.4 °F                                                                                                                         |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | No data available                                                                                                                       |
| <b>Lower</b>                            | No data available                                                                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                            |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Flammables area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Strong oxidizing agents.                                                                                                                                   |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component         | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH IDLH                                                                                                    | Mexico OEL (TWA)              |
|-------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------|
| Isopropyl alcohol | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> | TWA: 200 ppm<br>STEL: 400 ppm |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.  
Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                                  |
|----------------------------------------|--------------------------------------------------|
| Physical State                         | Liquid                                           |
| Appearance                             | Clear                                            |
| Odor                                   | No information available                         |
| Odor Threshold                         | No information available                         |
| pH                                     | No information available                         |
| Melting Point/Range                    | No data available                                |
| Boiling Point/Range                    | 150 - 152 °C / 302 - 305.6 °F @ 10 mmHg          |
| Flash Point                            | 28 °C / 82.4 °F                                  |
| Evaporation Rate                       | No information available                         |
| Flammability (solid,gas)               | Not applicable                                   |
| Flammability or explosive limits       |                                                  |
| Upper                                  | No data available                                |
| Lower                                  | No data available                                |
| Vapor Pressure                         | No information available                         |
| Vapor Density                          | 7.9                                              |
| Specific Gravity                       | 1.080                                            |
| Solubility                             | No information available                         |
| Partition coefficient; n-octanol/water | No data available                                |
| Autoignition Temperature               | No information available                         |
| Decomposition Temperature              | No information available                         |
| Viscosity                              | No information available                         |
| Molecular Formula                      | C <sub>8</sub> H <sub>20</sub> O <sub>4</sub> Ti |
| Molecular Weight                       | 228.15                                           |

## 10. Stability and reactivity

|                                         |                                                                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                                            |
| <b>Stability</b>                        | Moisture sensitive.                                                                                                                   |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                                                                               |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                                                               |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                         |

## 11. Toxicological information

### Acute Toxicity

|                            |                                                                               |
|----------------------------|-------------------------------------------------------------------------------|
| <b>Product Information</b> | No acute toxicity information is available for this product                   |
| <b>Oral LD50</b>           | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Dermal LD50</b>         | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Vapor LC50</b>          | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

### Component Information

| Component         | LD50 Oral                                  | LD50 Dermal         | LC50 Inhalation       |
|-------------------|--------------------------------------------|---------------------|-----------------------|
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat ) | 72.6 mg/L ( Rat ) 4 h |

|                                             |                          |
|---------------------------------------------|--------------------------|
| <b>Toxicologically Synergistic Products</b> | No information available |
|---------------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Irritating to eyes, respiratory system and skin Vapors may cause drowsiness and dizziness |
| <b>Sensitization</b>   | No information available                                                                  |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen.  |

| Component                  | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------|-----------|------------|------------|------------|------------|------------|
| Ethanol, titanium(4+) salt | 3087-36-3 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Isopropyl alcohol          | 67-63-0   | Not listed | Not listed | Not listed | Not listed | Not listed |

|                          |                          |
|--------------------------|--------------------------|
| <b>Mutagenic Effects</b> | No information available |
|--------------------------|--------------------------|

|                             |                           |
|-----------------------------|---------------------------|
| <b>Reproductive Effects</b> | No information available. |
|-----------------------------|---------------------------|

|                              |                           |
|------------------------------|---------------------------|
| <b>Developmental Effects</b> | No information available. |
|------------------------------|---------------------------|

|                       |                           |
|-----------------------|---------------------------|
| <b>Teratogenicity</b> | No information available. |
|-----------------------|---------------------------|

|                                 |                              |
|---------------------------------|------------------------------|
| <b>STOT - single exposure</b>   | Central nervous system (CNS) |
| <b>STOT - repeated exposure</b> | None known                   |

|                          |                          |
|--------------------------|--------------------------|
| <b>Aspiration hazard</b> | No information available |
|--------------------------|--------------------------|

|                                                   |                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
|---------------------------------------------------|-------------------------------------------------------------------------------------|

|                                        |                          |
|----------------------------------------|--------------------------|
| <b>Endocrine Disruptor Information</b> | No information available |
|----------------------------------------|--------------------------|

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| <b>Other Adverse Effects</b> | The toxicological properties have not been fully investigated. |
|------------------------------|----------------------------------------------------------------|

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

| Component         | Freshwater Algae                                                                                           | Freshwater Fish                                                                                                                                                                                                          | Microtox                                              | Water Flea                                      |
|-------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------|
| Isopropyl alcohol | EC50: > 1000 mg/L, 96h<br>(Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 72h<br>(Desmodesmus subspicatus) | LC50: = 9640 mg/L, 96h<br>flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h<br>(Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h<br>static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h<br>(Daphnia) | = 35390 mg/L EC50<br>Photobacterium phosphoreum 5 min | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Is not likely mobile in the environment.

| Component         | log Pow |
|-------------------|---------|
| Isopropyl alcohol | 0.05    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group III

### TDG

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group III

### IATA

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group III

### IMDG/IMO

UN-No UN1993  
 Proper Shipping Name Flammable liquid, n.o.s.  
 Hazard Class 3  
 Packing Group III

## 15. Regulatory information

### United States of America Inventory

| Component                  | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Ethanol, titanium(4+) salt | 3087-36-3 | X    | ACTIVE                                        | -                           |
| Isopropyl alcohol          | 67-63-0   | X    | ACTIVE                                        | -                           |

**Legend:****TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                  | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Ethanol, titanium(4+) salt | 3087-36-3 | X   | -   | 221-410-8 | X     | X    | X    | X    | X     | KE-13224 |
| Isopropyl alcohol          | 67-63-0   | X   | -   | 200-661-7 | X     | X    | X    | X    | X     | KE-29363 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

| Component         | CAS No  | Weight % | SARA 313 - Threshold Values % |
|-------------------|---------|----------|-------------------------------|
| Isopropyl alcohol | 67-63-0 | 5-15     | 1.0                           |

**SARA 311/312 Hazard Categories** See section 2 for more information**CWA (Clean Water Act)** Not applicable**Clean Air Act** Not applicable**OSHA** - Occupational Safety and Health Administration Not applicable**CERCLA** Not applicable**California Proposition 65** This product does not contain any Proposition 65 chemicals.**U.S. State Right-to-Know Regulations**

| Component         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------|---------------|------------|--------------|----------|--------------|
| Isopropyl alcohol | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.**Other International Regulations****Mexico - Grade** No information available**Authorisation/Restrictions according to EU REACH**

| Component | REACH (1907/2006) - Annex XIV - | REACH (1907/2006) - Annex XVII - | REACH Regulation (EC |
|-----------|---------------------------------|----------------------------------|----------------------|
|-----------|---------------------------------|----------------------------------|----------------------|

|                   | Substances Subject to Authorization | Restrictions on Certain Dangerous Substances                    | 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|-------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------|
| Isopropyl alcohol | -                                   | Use restricted. See item 75. (see link for restriction details) | -                                                                                |

<https://echa.europa.eu/substances-restricted-under-reach>

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                  | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Ethanol, titanium(4+) salt | 3087-36-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Isopropyl alcohol          | 67-63-0   | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                  | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Ethanol, titanium(4+) salt | 3087-36-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Isopropyl alcohol          | 67-63-0   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |

## 16. Other information

**Prepared By** Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

**Creation Date** 07-May-1997

**Revision Date** 25-Dec-2021

**Print Date** 25-Dec-2021

**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**